Cybersecurity & Ethical Hacking Lab
Task 1: Foundation & Environment Setup

Prepared by: Advay Sreejith
Date: April 2026
Lab Environment: Kali Linux + Metastploitabl2 + DVWA
Tools Used: Nmap · Wireshark· Burp Suite· Netcat· OpenSSL

This report documents the setup and verification of a professional ethical hacking lab environment, covering cybersecurity fundamentals, networking, cryptography, and tool familiarization

---

Table of Contents

1	Executive Summary	1
2	Lab Environment	2
3	Cybersecurity Fundamentals	3
4	Linux Fundamentals	5
5	Networking Basics	6
6	Cryptography	8
7	Tool Familiarization	10
8	Conclusion	13

1. Executive Summary

This report documents the completion of Task 1 of a 60-day Cybersecurity and Ethical Hacking program. The objective was to build a professional penetration testing lab from scratch and establish strong foundational knowledge across six domains: cybersecurity basics, lab environment setup, Linux fundamentals, networking, cryptography, and tool familiarization.

The lab environment consists of Kali Linux (192.168.189.131) as the attacker machine and Metasploitable2 (192.168.189.132) as the target, connected via an isolated VMware Host-Only network. DVWA (Damn Vulnerable Web App) runs natively on Metasploitable2 at http://192.168.189.132/dvwa.

All tools were verified operational including Nmap, Wireshark, Burp Suite, Netcat, and OpenSSL. An initial Nmap aggressive scan identified 23 open ports on Metasploitable2 with several critical vulnerabilities noted for exploitation in Task 4.

2. Lab Environment

2.1 Network Topology

The lab uses VMware Workstation with a Host-Only Adapter ensuring complete network isolation from the host machine

and the internet. This prevents any accidental exposure of vulnerable services.

Machine roles:
- Kali Linux 2024.x — Attacker — 192.168.189.131
- Metasploitable2 — Target — 192.168.189.132
- DVWA — Web App Target — 192.168.189.132/dvwa

2.2 Connectivity verification

Both machines were confirmed reachable via ICMP ping with 0% packet loss. ARP table confirmed Metasploitable2 MAC address as 00:0c:29:64:c6:81 — consistent with a VMware-assigned address.

3. Cybersecurity Fundamentals

3.1 CIA Triad
The CIA Triad forms the foundation of all information security practices. It defines three core principles that guide security decision-making:

Principle Violation	Definition	Real-World
Confidentiality	Data accessible only to authorized users	2017 Equifax breach - 147M SSNs exposed
Integrity	Data is accurate and unaltered	Stuxnet modified PLC code in Iranian nuclear centrifuges
Availability	Systems accessible when needed	2016 Dyn DDoS took down Twitter, Netflix, Github

3.2 Threat Types
The following threat types represent the most common attack categories encountered in real-world environments:

Threat	How it works	Real Incident
Pishing	Spoofed emails trick users into revealing credentials	2016 DNC hack via spear-phishing
Malware	Malicious software installed on victim system	WannaCry - 200K systems in 150 countries

DDoS	Flood target with traffic until it crashes	Mirai botnet attack on Dyn DNS, Oct 2016
SQL Injection	Malicious SQL inserted into input fields	2008 Heartland - 160M card numbers stolen
Brute Force	Systematically tries all possible passwords	2012 LinkedIn - 6.5M hatched passwords cracked
Ransomware	Encrypts files,demands payment for key	Colonial Pipeline 2021 - $4.4M ransom paid

3.3 Attack Vectors

Attack vectors are the pathways through which attackers gain unauthorized access to systems:

- Social Engineering — Manipulates humans rather than systems through pretexting, baiting, and tailgating. The human element is consistently the weakest link in any security chain.

- Wireless Attacks — Exploits Wi-Fi weaknesses including evil twin access points, WPA2 handshake capture, and deauthentication attacks.

- Insider Threats — Malicious or negligent actions by authorized users including data theft, credential sharing, and sabotage.

4. Linux Fundamentals

4.1 File System Navigation and Permissions

Linux file permissions follow a three-tier model: owner, group, and others. Each tier can have read (4), write (2), and execute (1) permissions. The chmod command modifies these permissions numerically or symbolically.

```
┌──(root㉿kali)-[~]
└─# ls -la
total 140
drwx------ 17 root root  4096 Mar 24 02:37 .
drwxr-xr-x 18 root root  4096 Mar 23 03:40 ..
-rw-r--r--  1 root root   220 Mar 23 03:40 .bash_logout
-rw-r--r--  1 root root  5551 Dec  2 21:31 .bashrc
-rw-r--r--  1 root root   607 Dec  2 21:31 .bashrc.original
drwx------  8 root root  4096 Mar 23 04:38 .cache
drwxr-xr-x 14 root root  4096 Mar 23 04:41 .config
drwxr-xr-x  2 root root  4096 Mar 23 03:40 Desktop
-rw-r--r--  1 root root    35 Mar 23 03:40 .dmrc
drwxr-xr-x  2 root root  4096 Mar 23 03:40 Documents
drwxr-xr-x  2 root root  4096 Mar 23 03:40 Downloads
-rw-r--r--  1 root root 11656 Dec  2 21:31 .face
lrwxrwxrwx  1 root root    11 Jan 19 09:30 .face.icon -> /root/.face
drwx------  3 root root  4096 Mar 23 03:40 .gnupg
-rw-------  1 root root     0 Mar 23 03:40 .ICEauthority
drwxr-xr-x  3 root root  4096 Mar 23 03:40 .java
drwxr-xr-x  5 root root  4096 Mar 23 03:40 .local
drwxr-xr-x  5 root root  4096 Mar 23 04:03 .mozilla
drwxr-xr-x  2 root root  4096 Mar 23 03:40 Music
drwxr-xr-x  2 root root  4096 Mar 23 03:56 Pictures
-rw-r--r--  1 root root   132 Nov 10 04:50 .profile
drwxr-xr-x  2 root root  4096 Mar 23 03:40 Public
drwx------  3 root root  4096 Mar 23 03:40 .ssh
drwxr-xr-x  2 root root  4096 Mar 23 03:40 Templates
drwxr-xr-x  2 root root  4096 Mar 23 03:40 Videos
-rw-------  1 root root    49 Mar 24 02:37 .Xauthority
-rw-------  1 root root  4327 Mar 24 02:37 .xsession-errors
-rw-------  1 root root  7130 Mar 23 03:56 .xsession-errors.old
-rw-r--r--  1 root root   336 Mar 23 03:40 .zprofile
-rw-r--r--  1 root root 10855 Dec  2 21:31 .zshrc
```

Figure 1: ls -la output showing file permissions in Kali Linux

```
┌──(root㉿kali)-[~]
└─# ls -la testfile.txt
-rw-r--r-- 1 root root 0 Mar 24 02:46 testfile.txt

┌──(root㉿kali)-[~]
└─# chmod 755 testfile.txt

┌──(root㉿kali)-[~]
└─# ls -la testfile.txt
-rwxr-xr-x 1 root root 0 Mar 24 02:46 testfile.txt

┌──(root㉿kali)-[~]
└─# chmod 600 testfile.txt

┌──(root㉿kali)-[~]
└─# ls -la testfile.txt
-rw------- 1 root root 0 Mar 24 02:46 testfile.txt
```

Figure 2: chmod demonstrating permission changes on a test file

4.2 Networking Commands

Key networking commands verified operational in the lab:
- ifconfig — displays network interface configuration
- ping — tests ICMP connectivity between hosts
- netstat -tuln — lists all listening ports and services
- traceroute — traces the network path to a destination

```
┌──(root㉿kali)-[~]
└─# netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:34595         0.0.0.0:*               LISTEN
```

Figure 3: netstat showing listening ports on Kali Linux

5. Networking Basics

5.1 OSI Model

The OSI model provides a conceptual framework for understanding how network communication works across seven distinct layers:

Layer	Name	Protocol Examples	Role
7	Application	HTTP, FTP, DNS, SMTP	User-facing services
6	Presentation	SSL/TLS, JPEG	Encryption and encoding
5	Session	NetBIOS, RPC	Session management
4	Transport	TCP, UDP	End-to-end delivery
3	Network	IP, ICMP, OSPF	Routing between networks
2	Data Link	Ethernet, ARP	Node-to-node delivery
1	Physical	Ethernet cable, WiFi	Raw bit transmission

5.2 TCP/IP and the Three-Way Handshake

TCP establishes connections using a three-way handshake before any data is exchanged:

- SYN — client sends synchronization request
- SYN-ACK — server acknowledges and responds
- ACK — client confirms, connection established

This was captured and verified in Wireshark during lab exercises.

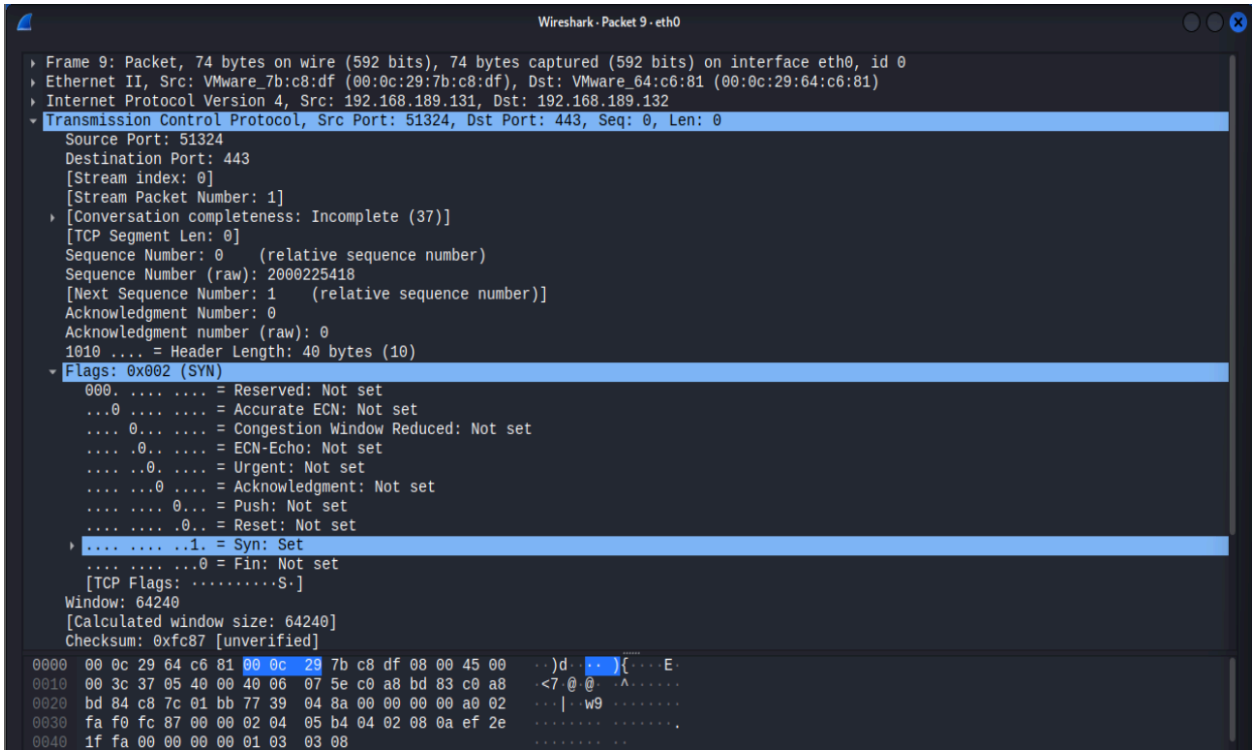


Figure 4: Wireshark capture showing TCP SYN flags during three-way handshake between Kali and Metasploitable2

5.3 DNS Resolution

DNS translates human-readable domain names into IP addresses. The dig command provides detailed DNS query information including TTL values, record types, and authoritative servers.

```
┌──(root㉿kali)-[~]
└─# dig google.com

; <<>> DiG 9.20.15-2-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62072
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             5       IN      A       142.250.66.14

;; Query time: 34 msec
;; SERVER: 192.168.189.2#53(192.168.189.2) (UDP)
;; WHEN: Tue Mar 24 03:01:09 EDT 2026
;; MSG SIZE  rcvd: 55
```

Figure 5: dig query showing DNS A record resolution with TTL and server information

5.4 HTTP Traffic Analysis

HTTP requests and responses were captured in Wireshark while browsing DVWA. Each packet demonstrates all OSI layers stacked from Physical through Application.

```
Wireshark · Packet 22 · eth0

▸ Frame 22: Packet, 415 bytes on wire (3320 bits), 415 bytes captured (3320 bits) on interface eth0, id 0
▸ Ethernet II, Src: VMware_7b:c8:df (00:0c:29:7b:c8:df), Dst: VMware_64:c6:81 (00:0c:29:64:c6:81)
▸ Internet Protocol Version 4, Src: 192.168.189.131, Dst: 192.168.189.132
▸ Transmission Control Protocol, Src Port: 58666, Dst Port: 80, Seq: 1, Ack: 1, Len: 349
▾ Hypertext Transfer Protocol
  ▾ GET /dvwa/login.php HTTP/1.1\r\n
      Request Method: GET
      Request URI: /dvwa/login.php
      Request Version: HTTP/1.1
    Host: 192.168.189.132\r\n
    User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
    Accept-Language: en-US,en;q=0.5\r\n
    Accept-Encoding: gzip, deflate\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    Priority: u=0, i\r\n
    \r\n
    [Response in frame: 26]
    [Full request URI: http://192.168.189.132/dvwa/login.php]
```

Figure 6: Wireshark HTTP capture showing OSI layers

in a single packet destined for DVWA

5.5 Network Reconnaissance

Traceroute confirmed a single-hop path between Kali and Metasploitable2, consistent with an isolated Host-Only network. nslookup verified DNS resolution against both the default resolver and Google's public DNS (8.8.8.8).

```
┌──(root㉿kali)-[~]
└─# traceroute 192.168.189.132
traceroute to 192.168.189.132 (192.168.189.132), 30 hops max, 60 byte packets
 1  * * *
 2  192.168.189.132 (192.168.189.132)  5.014 ms  4.900 ms  4.456 ms
```

Figure 7: traceroute confirming single-hop isolated lab network

```
┌──(root㉿kali)-[~]
└─# nslookup google.com
Server:         192.168.189.2
Address:        192.168.189.2#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.222.110
Name:   google.com
Address: 2404:6800:4007:804::200e
```

Figure 8: nslookup confirming DNS resolution via 192.168.189.2

6. Cryptography

6.1 Symmetric vs Asymmetric Encryption

Two fundamental encryption paradigms exist in modern cryptography:

- Symmetric encryption uses the same key for both encryption and decryption. It is fast and suitable for bulk data encryption. AES-256-CBC was demonstrated using OpenSSL.

- Asymmetric encryption uses a public/private key pair. The public key encrypts, the private key decrypts. RSA-2048 was demonstrated using OpenSSL pkeyutl.

6.2 Hashing and the Avalanche Effect

Hashing produces a fixed-length output from any input. A fundamental property is the avalanche effect — a single character change produces a completely different hash. This was demonstrated by hashing two nearly identical strings using MD5 and SHA256.

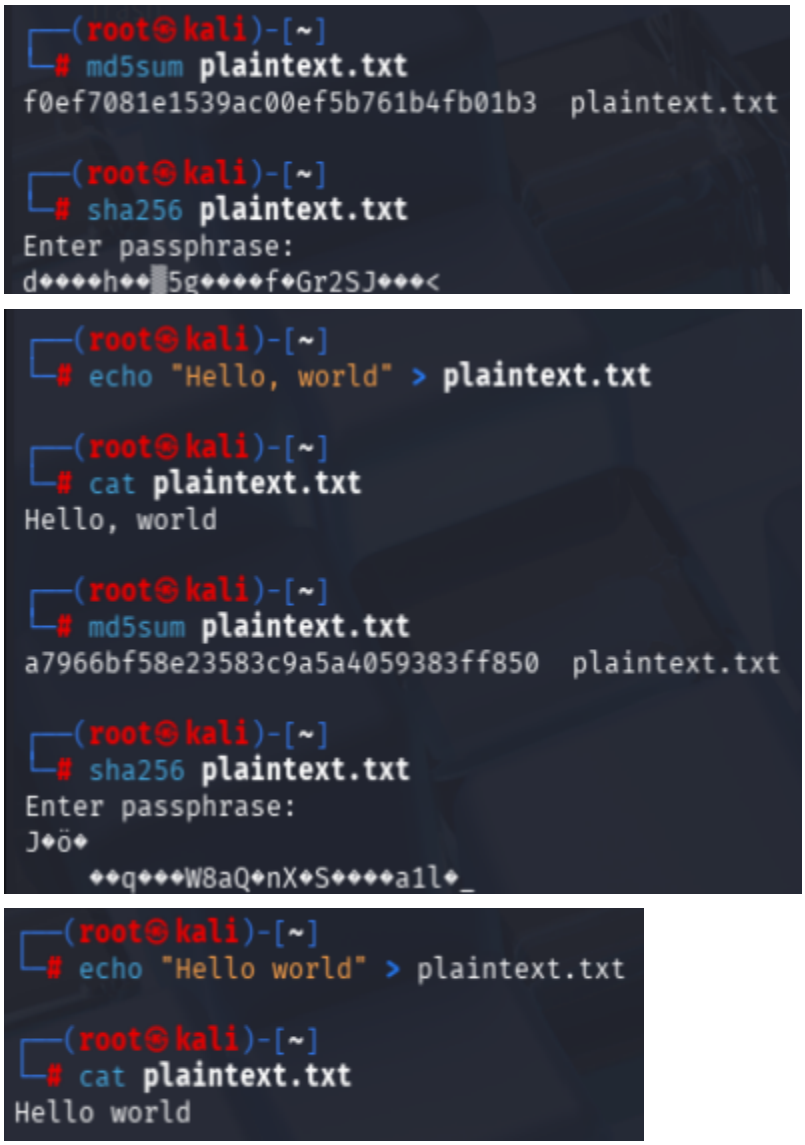


Figure 9: MD5 and SHA256 hashes demonstrating the avalanche effect — removing one character produces a completely different hash output

6.3 AES-256 Symmetric Encryption (OpenSSL)

A plaintext file was encrypted using AES-256-CBC with PBKDF2 key derivation and successfully decrypted back to the original message. The -pbkdf2 flag ensures modern key derivation rather than the deprecated legacy method.

```
┌──(root㉿kali)-[~]
└─# echo "Hello, Cybersecurity World" > plaintext.txt

┌──(root㉿kali)-[~]
└─# openssl enc -aes-256-cbc -salt -pbkdf2 -in plaintext.txt -out encrypted.enc
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:

┌──(root㉿kali)-[~]
└─# cat encrypted.enc
Salted__R◆c◆G◆l,
                S◆^◆8"◆.◆◆◆],~◆i◆◆◆◆
```

```
┌──(root㉿kali)-[~]
└─# openssl enc -aes-256-cbc -d -pbkdf2 -in encrypted.enc -out decrypted.txt

enter AES-256-CBC decryption password:

┌──(root㉿kali)-[~]
└─# cat decrypted.txt
Hello, Cybersecurity World
```

Figure 10: AES-256-CBC encryption producing binary ciphertext
Figure 11: Successful decryption recovering original plaintext

6.4 RSA Asymmetric Encryption (OpenSSL)

A 2048-bit RSA key pair was generated. A message was encrypted using the public key and decrypted using the private key, confirming asymmetric encryption works correctly. Note: rsautl was deprecated in OpenSSL 3.0 — pkeyutl was used instead.

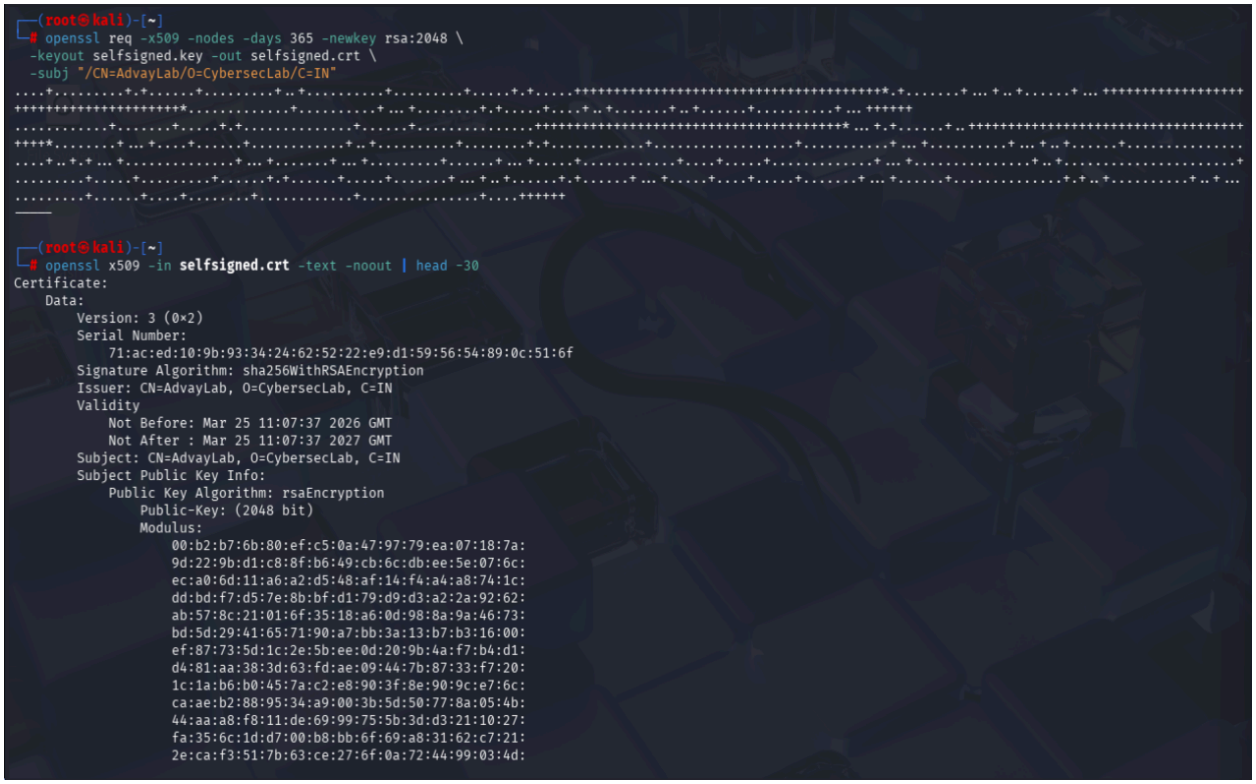
```
┌──(root㉿kali)-[~]
└─# openssl genrsa -out private.key 2048

┌──(root㉿kali)-[~]
└─# openssl rsa -in private.key -pubout -out public.key
writing RSA key

┌──(root㉿kali)-[~]
└─# cat public.key
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAyA8uDHwHYDySCjRkP2/n
BNLpHBZhhT1ApTb1IJYdBRR0iXrJd3JRUzNJUuHHq9vTQBJG3ydJe0EllmLiiX3x
ifABaMhmGCYHu9DzCaKUXtY771wDwFt64T+lkS2Q/VAlpQQJiT3Lz+KE5bxFLRna
7x8IeDNLoxiR1jiYZa3YxU1jh1hDR3dhtwgzN6Vb455oIDjg5ZI1QlgIZOEj18eH
Ozwj6vM2+cyLjVSTtGYbzfOZT8GAU2J9bkUrslTEMtkHE+cSKiXtKOR6aArJmla2
5xspxhgRNVCFKWlxeYaldzBSuQ0bp3hPR7WKjLzawTSD7J37m/2zms3ZlKH6RZno
bwIDAQAB
-----END PUBLIC KEY-----
```

```
┌──(root㉿kali)-[~]
└─# echo "Secret message for RSA demo" > rsa_plain.txt

┌──(root㉿kali)-[~]
└─# openssl pkeyutl -encrypt -inkey public.key -pubin -in rsa_plain.txt -out rsa_encrypted.bin

┌──(root㉿kali)-[~]
└─# openssl pkeyutl -decrypt -inkey private.key -in rsa_encrypted.bin -out rsa_decrypted.txt

┌──(root㉿kali)-[~]
└─# cat rsa_decrypted.txt
Secret message for RSA demo
```

Figure 12: RSA 2048-bit key pair generation and public key output
Figure 13: RSA encryption and decryption using pkeyutl

6.5 Self-Signed Certificate

A self-signed X.509 certificate was generated for AdvayLab/CybersecLab valid for 365 days. This demonstrates the certificate structure used in SSL/TLS — issuer, validity period, subject, and public key details.



```
┌──(root㉿kali)-[~]
└─# openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout selfsigned.key -out selfsigned.crt \
  -subj "/CN=AdvayLab/O=CybersecLab/C=IN"
....+.........+.+......+.........+..+..........+.........+.....+.+.....++++++++++++++++++++++++++++++++++++++++++*.+.......+ ... + .. +......+ ... ++++++++++++++++++
++++++++++++++++++++*............+..........+ ... +........+.+.....+....+ .. +.......+ .. +......+..........+ ... ++++++
.............+.......+....+.+..............+......+..............++++++++++++++++++++++++++++++++++++++++++ ... +.+......+ .. ++++++++++++++++++++++++++++++++++++
+++++........+ ... +....+......+............+ .. +..........+........+.+............+..................+............+ ... +...........+ ... + .. +......+..............
....+ .. +.+ ... +..............+ ... +.......+ ... +.........+......+ ... +.....+............+....+.....+............+ ... +..............+ .. +......................+
.........+.....+.........+......+.+......+.....+.......+ ... + .. +......+.+......+ ... +.....+....+.....+.......+ ... +......+..............+.+ .. +..........+ .. + ...
.........+......+....+........+............+..............+....++++++
────
```

```
┌──(root㉿kali)-[~]
└─# openssl x509 -in selfsigned.crt -text -noout | head -30
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            71:ac:ed:10:9b:93:34:24:62:52:22:e9:d1:59:56:54:89:0c:51:6f
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: CN=AdvayLab, O=CybersecLab, C=IN
        Validity
            Not Before: Mar 25 11:07:37 2026 GMT
            Not After : Mar 25 11:07:37 2027 GMT
        Subject: CN=AdvayLab, O=CybersecLab, C=IN
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:b2:b7:6b:80:ef:c5:0a:47:97:79:ea:07:18:7a:
                    9d:22:9b:d1:c8:8f:b6:49:cb:6c:db:ee:5e:07:6c:
                    ec:a0:6d:11:a6:a2:d5:48:af:14:f4:a4:a8:74:1c:
                    dd:bd:f7:d5:7e:8b:bf:d1:79:d9:d3:a2:2a:92:62:
                    ab:57:8c:21:01:6f:35:18:a6:0d:98:8a:9a:46:73:
                    bd:5d:29:41:65:71:90:a7:bb:3a:13:b7:b3:16:00:
                    ef:87:73:5d:1c:2e:5b:ee:0d:20:9b:4a:f7:b4:d1:
                    d4:81:aa:38:3d:63:fd:ae:09:44:7b:87:33:f7:20:
                    1c:1a:b6:b0:45:7a:c2:e8:90:3f:8e:90:9c:e7:6c:
                    ca:ae:b2:88:95:34:a9:00:3b:5d:50:77:8a:05:4b:
                    44:aa:a8:f8:11:de:69:99:75:5b:3d:d3:21:10:27:
                    fa:35:6c:1d:d7:00:b8:bb:6f:69:a8:31:62:c7:21:
                    2e:ca:f3:51:7b:63:ce:27:6f:0a:72:44:99:03:4d:
```

Figure 14: Self-signed X.509 certificate details showing issuer, validity period, and RSA public key

7. Tool Familiarization

7.1 Nmap — Network Scanner

Nmap was used to perform comprehensive reconnaissance against Metasploitable2. Four scan types were executed:

- SYN Scan (-sS) — stealthy half-open scan identifying 23 open TCP ports
- Service Version Detection (-sV) — identified exact service versions running on each port
- OS Detection (-O) — fingerprinted target OS as Linux 2.6.9-2.6.33
- Aggressive Scan (-A) — combined OS detection, version detection, script scanning and traceroute

```
┌──(root㉿kali)-[~]
└─# nmap -sS 192.168.189.132
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 07:10 -0400
Nmap scan report for 192.168.189.132
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:64:C6:81 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.92 seconds
```

Figure 15: Nmap SYN scan revealing 23 open ports on Metasploitable2

```
┌──(root㉿kali)-[~]
└─# nmap -sV 192.168.189.132
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 07:10 -0400
Nmap scan report for 192.168.189.132
Host is up (0.0019s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13?
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:64:C6:81 (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 148.31 seconds
```

Figure 16: Nmap service version detection identifying exact versions of running services

```
┌──(root㉿kali)-[~]
└─# nmap -O 192.168.189.132
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 07:18 -0400
Nmap scan report for 192.168.189.132
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:64:C6:81 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.18 seconds
```

Figure 17: Nmap OS detection fingerprinting target as Linux 2.6.9-2.6.33

7.2 Critical Vulnerabilities Identified

The Nmap aggressive scan revealed several critical vulnerabilities on Metasploitable2 that will be exploited in Task 4:

Port	Service	Version	Risk
21	FTP	vsftp 2.3.4	CRITICAL-CVE-2011-2523 backdoor
1524	Bindshell	Root shell	CRITICAL — open root shell
6667	IRC	UnrealIRCd 3.2.8.1	CRITICAL — CVE-2010-2075 backdoor
23	Telnet	Linux telnetd	HIGH — plaintext protocol
3306	MySQL	5.0.51a	HIGH — outdated version

7.3 Wireshark — Packet Analyzer

Wireshark was used to capture and analyze three types of network traffic:

- ICMP — ping packets between Kali and Metasploitable2
- TCP — three-way handshake showing SYN/SYN-ACK/ACK flags
- HTTP — full request to DVWA showing all OSI layers
- DNS — query and response showing A record resolution

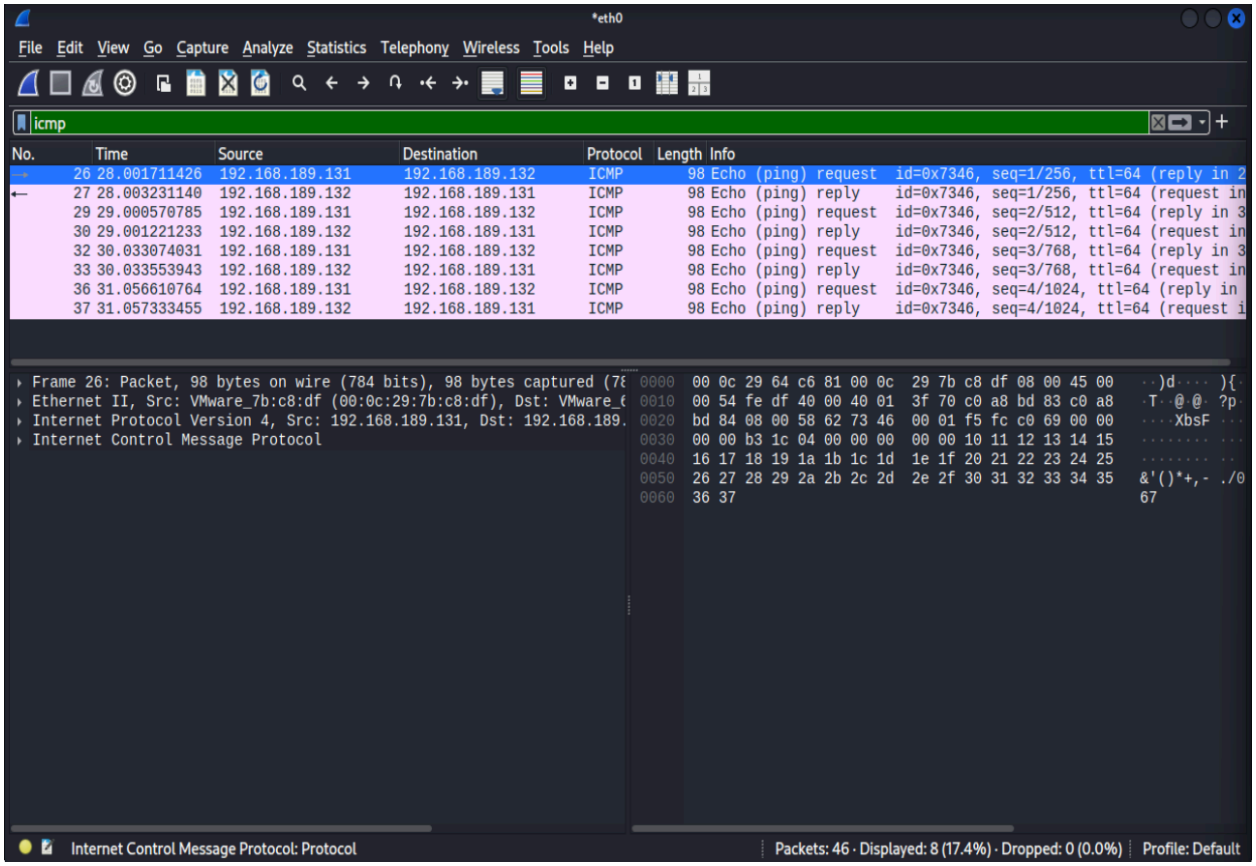


Figure 18: Wireshark ICMP capture showing ping request/reply pairs between lab machines

7.4 Burp Suite — Web Proxy

Burp Suite was configured as a proxy on 127.0.0.1:8080. Firefox was configured to route traffic through Burp, allowing interception and inspection of HTTP requests to DVWA. The Proxy → HTTP History tab confirmed all requests were being captured successfully.

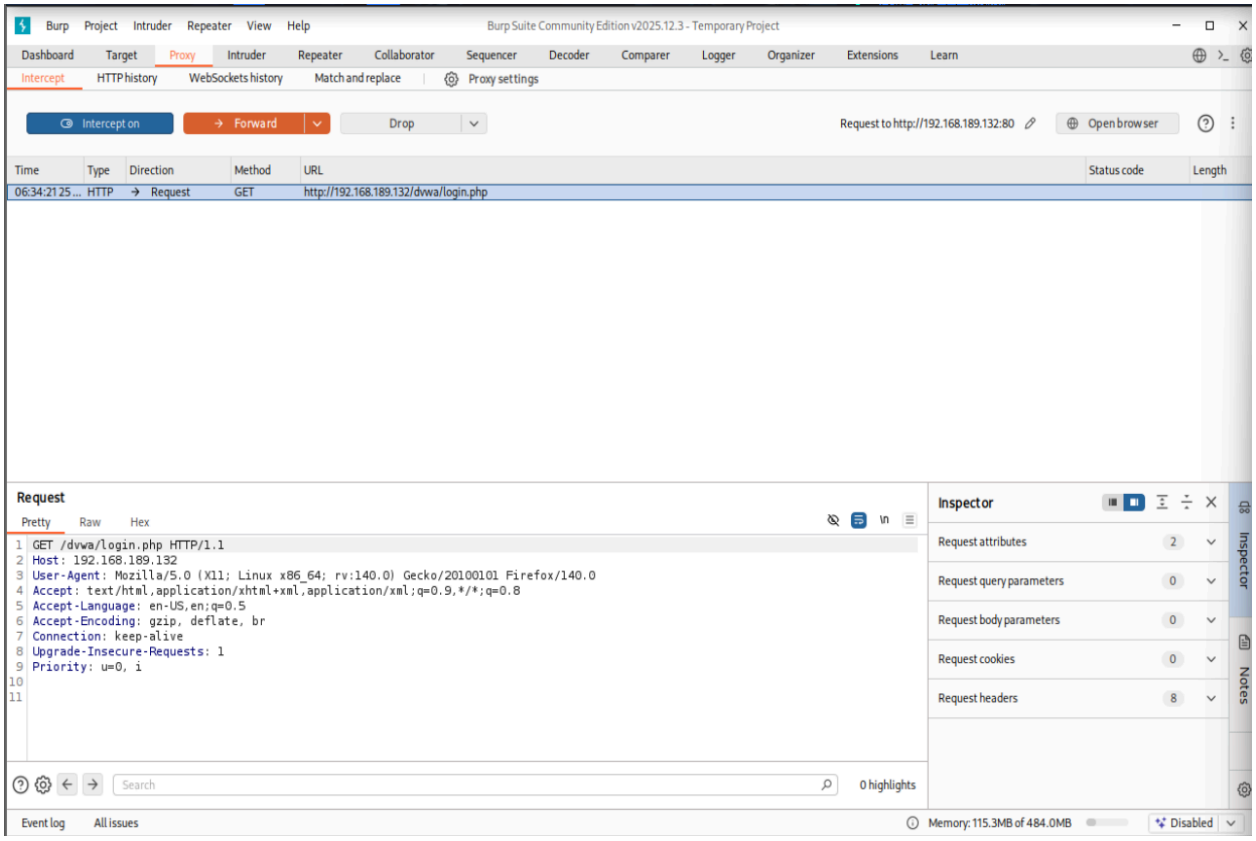


Figure 19: Burp Suite intercepting raw HTTP request to DVWA '

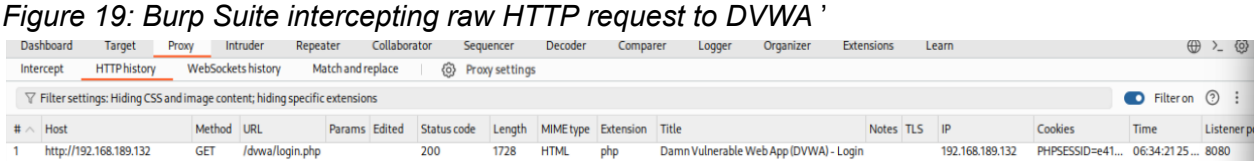


Figure 20: Burp Suite HTTP History showing captured requests

7.5 Netcat — Network Utility

Netcat was demonstrated as both a listener and client:

- TCP listener established on port 4444
- Bidirectional message transfer confirmed
- File transfer demonstrated via piped input/output

Netcat forms the basis of reverse shell communication used extensively in Task 4 exploitation exercises.

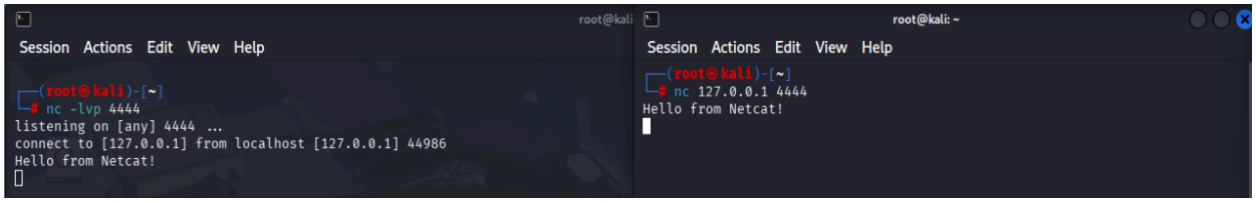


Figure 21: Netcat listener receiving message on port 4444

Figure 22: Netcat file transfer confirmation

8. Conclusion

Task 1 successfully established a fully operational penetration testing lab and documented foundational knowledge across all six required domains.

Key achievements:

- Lab environment configured with isolated Host-Only networking — zero connectivity issues throughout

- All five tools verified operational: Nmap, Wireshark, Burp Suite, Netcat, and OpenSSL

- Cryptographic operations demonstrated hands-on: AES-256 symmetric encryption, RSA-2048 asymmetric encryption, SHA256 hashing, and X.509 certificate generation

- Initial reconnaissance of Metasploitable2 identified 23 open ports including three critical backdoor

vulnerabilities (vsftpd 2.3.4, UnrealIRCd 3.2.8.1, open bindshell on port 1524) — documented for exploitation in Task 4

- All findings documented with evidence in a public GitHub repository at: https://github.com/adv003/cybersec-ethical-hacking-lab

The lab is fully prepared for Task 2 — Network Security and Scanning — beginning with active reconnaissance and vulnerability scanning against Metasploitable2.